

Recall from last class:

- Let $A = [\vec{a}_1 \ \dots \ \vec{a}_n]$ be an m by n matrix. We express A by

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

where a_{ij} is the entry in the i -th row and j -th column.

- **Theorem:** Let A , B , and C be matrices of the same size, and let r and s be real numbers. Then the following properties hold:

1. $A + B = B + A$.
2. $(A + B) + C = A + (B + C)$.
3. $A + 0 = A$.
4. $r(A + B) = rA + rB$.
5. $(r + s)A = rA + sA$.
6. $r(sA) = rsA$.

- If A is a $m \times n$ matrix and B is an $n \times p$ matrix with columns $\vec{b}_1, \dots, \vec{b}_p$, then AB is the $m \times p$ matrix defined by

$$AB = A[\vec{b}_1 \ \dots \ \vec{b}_p] = [A\vec{b}_1 \ \dots \ A\vec{b}_p].$$

- Note

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}.$$

- **Theorem:**

1. $A(BC) = (AB)C$.
2. $A(B + C) = AB + AC$.
3. $(B + C)A = BA + CA$.
4. $r(AB) = (rA)B = A(rB)$.
5. $I_m A = A = A I_n$.

- Given an $m \times n$ matrix A , then the **transpose** of A , denoted A^T is the $n \times m$ matrix given by taking the columns of A as rows.

- **Theorem:** Let A be an m by n matrix, B be an n by p matrix, and $r \in \mathbb{R}$.

1. $(A^T)^T = A$.
2. $(A + B)^T = A^T + B^T$.
3. $(rA)^T = rA^T$.
4. $(AB)^T = B^T A^T$.

Definition 1. An n by n matrix A is **invertible** if there is an n by n matrix C so that $AC = I_n$ and $CA = I_n$. We denote the inverse of A by A^{-1} .

Natural Questions:

1. Do inverses always exist?
2. If not, when do they exist?
3. If they exist, are they unique?
4. How can we find them?

1. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Determine when A is invertible and find A^{-1} .

Theorem: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Definition 2. We define the **determinant** of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, denoted by $\det(A)$, by

$$\det(A) = ad - bc.$$

Theorem: If A is an n by n invertible matrix, then for each $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ has a unique solution; namely $\vec{x} = A^{-1}\vec{b}$.

2. Prove the Theorem.

Theorem: Suppose A and B are n by n invertible matrices, then the following hold.

(a) A^{-1} is invertible and $(A^{-1})^{-1} = A$.

(b) $(AB)^{-1} = B^{-1}A^{-1}$.

(c) $(A^T)^{-1} = (A^{-1})^T$.

3. Prove the Theorem.

Definition 3. *An elementary matrix E is a matrix obtained from the identity matrix by a single elementary row operation.*

4. Give some examples of elementary matrices.

Theorem: A is invertible if and only if A is row equivalent to the identity matrix. Furthermore, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .

5. Prove the theorem.

6.

(a) Let $A = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 5 & 6 \\ 5 & -4 & 5 \end{bmatrix}$. Find A^{-1} if possible.

(b) Let $A = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$. Find A^{-1} if possible.

Invertible Matrix Theorem Let A be an $n \times n$ matrix, then the following are all equivalent.

- (a) A is invertible.
- (b) A is row-equivalent to I_n .
- (c) A has n pivot positions.
- (d) $A\vec{x} = \vec{0}$ has only the trivial solution.
- (e) The columns of A are linearly independent.
- (f) $x \mapsto A\vec{x}$ is one-to-one (injective).
- (g) For all $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ is consistent.
- (h) The columns of A span $\mathbb{R}^n$.
- (i) $\vec{x} \mapsto A\vec{x}$ is onto (surjective).
- (j) There exists a matrix C such that $CA = I_n$.
- (k) There exists a matrix D such that $AD = I_n$.
- (l) A^T is invertible.

7. Prove the theorem.

8. Determine if the following matrices are invertible.

(a) Let $A = \begin{bmatrix} 3 & 0 & 0 \\ -3 & -4 & 0 \\ 8 & 5 & -3 \end{bmatrix}$.

(b) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

Theorem: Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be linear and let A be the associated matrix. Then T is invertible if and only if A is invertible.

9. Prove the theorem.

Inverses – Answers / Solutions

1. We want to find e, f, g and h such that

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Thus we get the following system of equations:

$$\begin{aligned} ae + gb &= 1 \\ af + bh &= 0 \\ ce + dg &= 0 \\ cf + dh &= 1 \end{aligned}$$

Multiplying the first equation by c and the third equation by $-a$ and adding them together yields: $bcg - adg = c$, and assuming $ad - bc \neq 0$, we get $g = \frac{-c}{ad-bc}$.

Continuing in a similar manner we get $e = \frac{d}{ad-bc}$, $h = \frac{a}{ad-bc}$, and $f = \frac{-b}{ad-bc}$.

Thus the matrix is invertible if and only if $ad - bc \neq 0$, and we have an explicit formula for the inverse matrix.

2. Since A is invertible, we know A^{-1} exists. Let $\vec{x} = A^{-1}\vec{b}$, then we check $\vec{x}$ is a solution. Here $A\vec{x} = AA^{-1}\vec{b} = \vec{b}$, and hence $\vec{x}$ is a solution.

Now we check it is the unique solution. Suppose $\vec{u}$ is another solution, then $A\vec{u} = \vec{b}$. Since A is invertible, we can multiply both sides by A^{-1} to get $A^{-1}A\vec{u} = A^{-1}\vec{b}$, and hence $\vec{u} = A^{-1}\vec{b} = \vec{x}$. Thus $\vec{x}$ is the only solution.

3. (a) Suppose A is invertible. We check that A is the inverse of A^{-1} ; namely, $AA^{-1} = I_n$ and $A^{-1}A = I_n$. Thus the inverse of A^{-1} is A ; in symbols $(A^{-1})^{-1} = A$.
- (b) Again suppose A and B are both invertible matrices. We check that $B^{-1}A^{-1}$ is the inverse of AB : $ABB^{-1}A^{-1} = I_n$ and $B^{-1}A^{-1}AB = I_n$. Thus by the definition of inverse matrix, $B^{-1}A^{-1}$ is the inverse of AB ; i.e. $(AB)^{-1} = B^{-1}A^{-1}$.
- (c) Suppose A is an invertible matrix. We want to prove that the inverse of A^T is $(A^{-1})^T$. Again we check the conditions in the definition of matrix inverse: $A^T(A^{-1})^T = (A^{-1}A)^T = (I_n)^T = I_n$ and $(A^{-1})^T A^T = (AA^{-1})^T = (I_n)^T = I_n$ where we used $(AB)^T = B^T A^T$. Thus $(A^T)^{-1} = (A^{-1})^T$.

4. Let

$$E_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, E_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, E_3 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}.$$

Now E_1 corresponds to scaling the first row by 2, E_2 corresponds to interchanging the first two rows, and E_3 corresponds to adding two times the first row to the second row. Notice each elementary matrix is invertible by undoing the corresponding elementary row operation. Thus

$$E_1^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix}, E_2^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, E_3^{-1} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}.$$

Note:

- Elementary matrices are invertible.
- If E is an elementary matrix and A is a matrix, then EA corresponds to performing E 's row operation on A .

5. *Proof.* We prove both implications directly. Let A be an n by n matrix.

$\implies$ Suppose A is invertible. We previously proved that for all $\vec{b} \in \mathbb{R}^n$, the matrix equation $A\vec{x} = \vec{b}$ is consistent. We proved in chapter one that the matrix equation $A\vec{x} = \vec{b}$ is consistent if and only if A has a pivot position in each row. Since A is an n by n matrix, we know there is also a pivot position in each column. Thus the reduced row echelon form of A is I_n , and hence A is row equivalent to I_n .

$\impliedby$ Suppose A is row equivalent to I_n . Thus there exists elementary matrices $E_1, \dots, E_p$ such that

$$A \sim E_1 A \sim E_2 E_1 A \sim \dots \sim E_p E_{p-1} \dots E_2 E_1 A = I_n.$$

Since elementary matrices are invertible and the product of invertible matrices is invertible, we get A is invertible and $A^{-1} = E_p \dots E_2 E_1$. Thus, if we apply the same sequence of row operations that transforms A to I_n to the identity matrix we get $E_p \dots E_2 E_1 I_n = A^{-1}$. $\square$

6. (a) Let $A = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 5 & 6 \\ 5 & -4 & 5 \end{bmatrix}$. We try augment A by I_n and try to use the result that if A is invertible, then $[A|I_n] \sim [I_n|A^{-1}]$. Thus

$$\left(\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ -1 & 5 & 6 & 0 & 1 & 0 \\ 5 & -4 & 5 & 0 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 1 & 1 & 0 \\ 0 & 6 & 10 & -5 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 1 & 1 & 0 \\ 0 & 0 & 0 & -7 & -2 & 1 \end{array} \right),$$

and now we're stuck. Since A is not row-equivalent to I_n (which we're seeing on left), the algorithm fails to find an inverse for A . Thus we know A is not invertible.

- (b) Let $A = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$. We again augment A by I_n and try to use the result that if A is invertible, then $[A|I_n] \sim [I_n|A^{-1}]$. Thus

$$\begin{aligned} \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{array} \right) &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & -3 & 8 & -2 & 0 & 1 \end{array} \right) \\ &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{array} \right) \\ &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 2 & \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{array} \right), \end{aligned}$$

$$\text{and hence } A^{-1} = \begin{bmatrix} 8 & 3 & 1 \\ 10 & 4 & 1 \\ \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{bmatrix}.$$

7. *Proof.* There are many ways to organize all the implications. Here I try to use previous results from class as much as possible.

(a) $\implies$ (j) by definition.

(j) $\implies$ (d) by Theorem 2.5 (or above on this handout).

(d) $\implies$ (c) by Theorem 1.4.

(c) $\implies$ (b) since A is square.

(b) $\implies$ (a) by Theorem 2.7 (or above on this handout).

Thus we have established (a) $\iff$ (b) $\iff$ (c) $\iff$ (d) $\iff$ (j).

Note (a) $\implies$ (k) by definition of invertibility. We prove (k) $\implies$ (g). Suppose D is an $n \times n$ matrix such that $AD = I_n$. Let $\vec{b} \in \mathbb{R}^n$, and define $\vec{y} = D\vec{b}$. Now $A\vec{y} = AD\vec{b} = I_n\vec{b} = \vec{b}$, and hence $\vec{y}$ is a solution to the matrix equation $A\vec{x} = \vec{b}$. Thus the matrix equation is always consistent, and hence (g) follows. Now we prove (g) $\implies$ (a). By Theorem 1.4, $A\vec{x} = \vec{b}$ being consistent for any $\vec{b}$ is equivalent to A having a pivot position in each row. Since A is square, we know A must be row equivalent to the identity matrix. By the theorem above, A is invertible, and hence (a) holds.

Finally note (d) $\iff$ (e) $\iff$ (f) and (g) $\iff$ (h) $\iff$ (i) follows from Theorem 1.12. Finally we prove (a) $\iff$ (l) above on this handout. $\square$

8. (a) We could row-reduce A to determine if A is row equivalent to the identity matrix, but in this case we can see the columns of A are linearly independent, and hence we know A must be invertible from the invertible matrix theorem.

(b) We have previously seen this matrix does not span all of $\mathbb{R}^3$, and hence we can use the invertible matrix theorem to conclude A is not invertible.

9. *Proof.* We prove both implications directly from the definitions.

$\implies$ Suppose T is invertible, then there exists a function $T^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $T(T^{-1}(\vec{x})) = \vec{x}$ and $T^{-1}(T(\vec{x})) = \vec{x}$ for all $\vec{x} \in \mathbb{R}^n$. If we knew that T^{-1} was linear, then it would have an associated matrix we could use.

Claim: T^{-1} is linear.

Proof of Claim: Let $\vec{x}, \vec{y} \in \mathbb{R}^n$ and $c \in \mathbb{R}$. Since T is invertible, it is bijective. Thus there exists unique $\vec{a}, \vec{b} \in \mathbb{R}^n$ such that $T(\vec{a}) = \vec{x}$ and $T(\vec{b}) = \vec{y}$. Now we check the conditions for linearity:

$$T^{-1}(\vec{x} + \vec{y}) = T^{-1}(T(\vec{a}) + T(\vec{b})) = T^{-1}(T(\vec{a} + \vec{b})) = \vec{a} + \vec{b} = T^{-1}(\vec{x}) + T^{-1}(\vec{y}).$$

and

$$T^{-1}(c\vec{x}) = T^{-1}(cT(\vec{a})) = T^{-1}(T(c\vec{a})) = c\vec{a} = cT^{-1}(\vec{x}).$$

Thus T^{-1} is linear.

Since T^{-1} is linear, it has an associated matrix D . Now

$$AD\vec{x} = T(T^{-1}(\vec{x})) = \vec{x}$$

and

$$DA\vec{x} = T^{-1}(T(\vec{x})) = \vec{x}.$$

Thus D is the inverse matrix of A , and hence A is invertible.

$\Leftarrow$ Suppose A is invertible. Define $S : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $S(\vec{x}) = A^{-1}\vec{x}$. Then $T(S(\vec{x})) = AA^{-1}\vec{x} = \vec{x}$ and $S(T(\vec{x})) = A^{-1}A\vec{x} = \vec{x}$. Thus S is the inverse function of T , and hence T is invertible. $\square$